Chapter 25

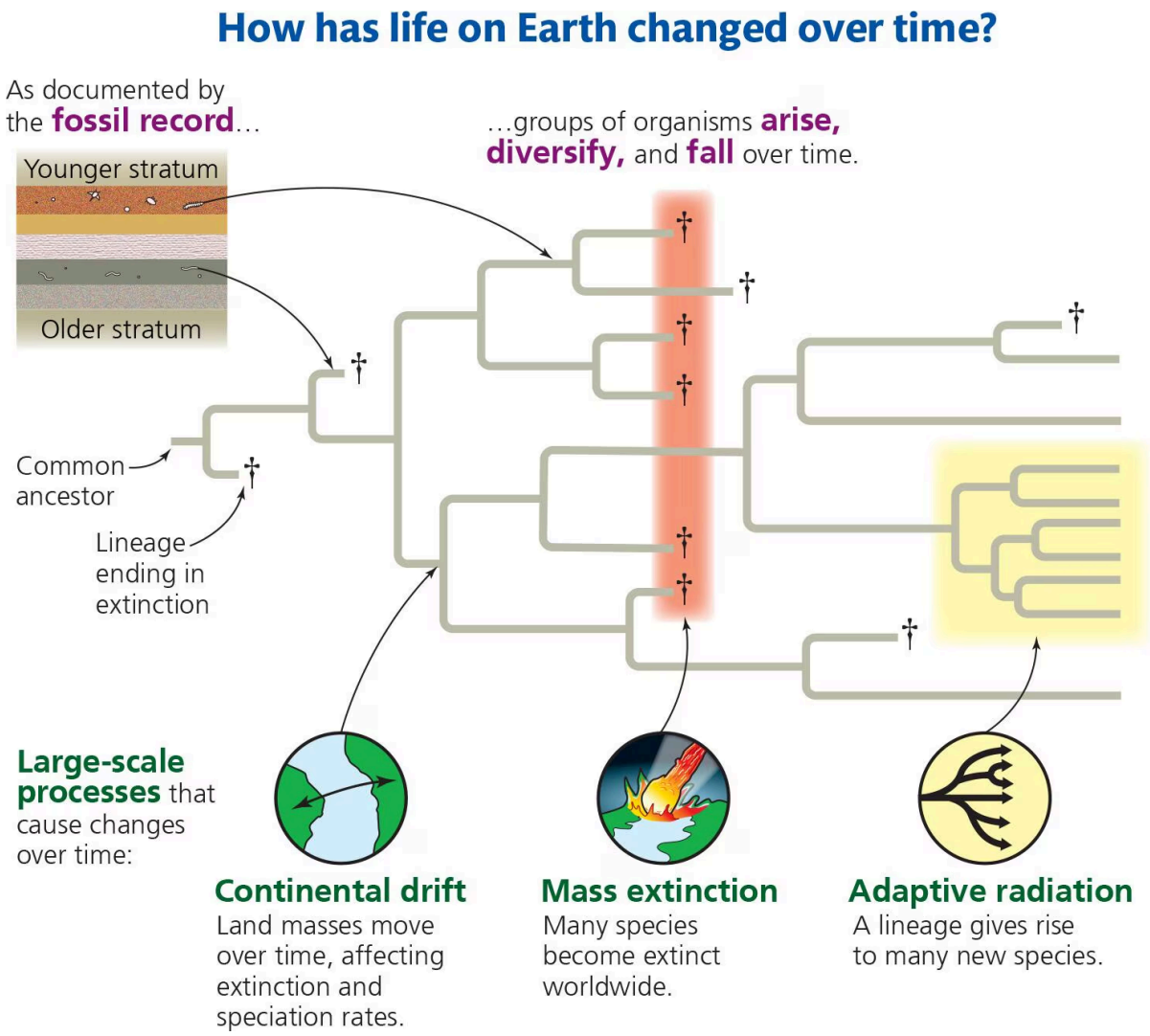
Campbell Biology, AP Edition, 12e

Chapter 25

The History of Life on Earth

Figure 25.1

In the 1870s, researchers working in the searing heat of the Sahara Desert were startled to discover fossils of ancient whales. These fossils were spectacular not only for where they were found, but also for documenting the transition from life on land to life in the sea—just one example of the sweeping changes seen in the history of life on Earth.



As documented in the fossil record, groups of organisms arise, diversify, and fall over time. Older fossils are found in older stratum. Large-scale processes that cause changes over time include the following: continent drift, where land masses move over time, affecting extinction and speciation rates; mass extinction, where many species become extinct worldwide; and adaptive radiation, where a lineage gives rise to many new species.

Key Concepts

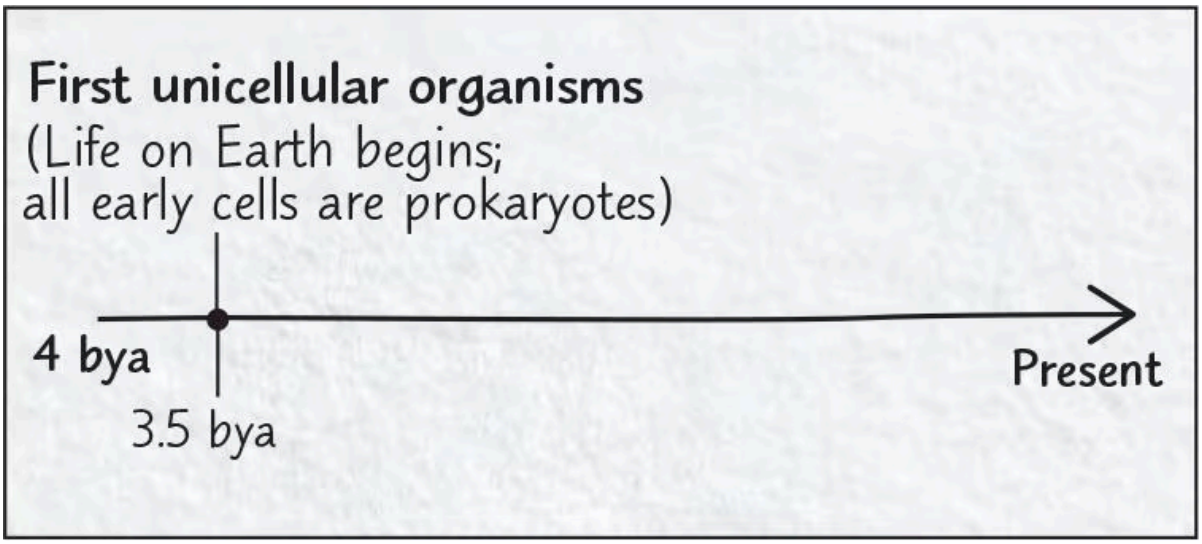
- 25.1**Conditions on early Earth made the origin of life possible**
- 25.2**The fossil record documents the history of life**
- 25.3**Key events in life's history include the origins of unicellular and multicellular organisms and the colonization of land**
- 25.4**The rise and fall of groups of organisms reflect differences in speciation and extinction rates**

- 25.5**Major changes in body form can result from changes in the sequences and regulation of developmental genes**
- 25.6**Evolution is not goal oriented**

Speciation and extinction have occurred throughout Earth's history **(BIG IDEA 1)**, with changes in the conditions on Earth leading to divergence in homeostatic **(BIG IDEA 2)** and developmental mechanisms **(BIG IDEA 3)**.

Study Tip

Draw a time line: Draw a time line like the example shown here, and mark when the following key evolutionary events occurred: "oxygen revolution," origin of eukaryotes, first multicellular organisms, colonization of land by large eukaryotes, origin of tetrapods, and the Permian mass extinction. Briefly describe the significance of each event. You can also add more events to your time line as you study.



The whale fossils discovered in the Sahara Desert (see Figure 25.1) are one example of evidence that past organisms were very different from those presently living. The sweeping changes in life on Earth as revealed by such fossils illustrate **macroevolution**, the broad pattern of evolution above the species level. Examples of macroevolutionary change include the emergence of terrestrial vertebrates through a series of speciation events, the impact of mass extinctions on biodiversity, and the origin of key adaptations such as flight.

Taken together, such changes provide a grand view of the evolutionary history of life. We'll examine that history in this chapter, beginning with a discussion of hypotheses regarding the origin of life. This is the most speculative topic of the entire unit, because no fossil evidence of that seminal episode exists.

Get Ready for This Chapter

In this chapter, you will learn about the history of life on Earth, focusing on evidence from the fossil record and mechanisms that have shaped the rise and fall of different groups of organisms over time. Before beginning this chapter, you should be able to do the following:

- Distinguish between prokaryotic and eukaryotic organisms (see Concept 6.2, Figure 6.5, and Figure 6.8).
- Explain the process of speciation (see Concept 24.2, Figure 24.5).
- Interpret an evolutionary tree (see Concept 22.3, Figure 22.17).

The following questions provide a quick check of your basic knowledge in these areas.

1. Which of the following is found in eukaryotic cells but not in prokaryotic cells?

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DNA
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That's incorrect. Try again.

Incorrect. Review the following material: A cell wall, DNA, and a plasma membrane can all be found in prokaryotic as well as eukaryotic cells.

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cell wall
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That's incorrect. Try again.

Incorrect. Review the following material: A cell wall, DNA, and a plasma membrane can all be found in prokaryotic as well as eukaryotic cells.

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membrane-enclosed organelles
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That's correct.

Correct! If you had trouble with this question, review the following material: A cell wall, DNA, and a plasma membrane can all be found in prokaryotic as well as eukaryotic cells. But only eukaryotes have membrane-enclosed organelles such as a true nucleus, mitochondria, and chloroplasts.

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plasma membrane
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That's incorrect. Try again.

Incorrect. Review the following material: A cell wall, DNA, and a plasma membrane can all be found in prokaryotic as well as eukaryotic cells.

2. Which of the following can promote allopatric speciation but *not* sympatric speciation?

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natural selection
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That's incorrect. Try again.

Incorrect. Review the following material: Allopatric speciation occurs when a population becomes geographically isolated from other members of its species; in contrast, sympatric speciation occurs in populations that are not geographically isolated.

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polyploidy
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That's incorrect. Try again.

Incorrect. Review the following material: Allopatric speciation occurs when a population becomes geographically isolated from other members of its species; in contrast, sympatric speciation occurs in populations that are not geographically isolated.

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        sexual selection
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That's incorrect. Try again.

Incorrect. Review the following material: Allopatric speciation occurs when a population becomes geographically isolated from other members of its species; in contrast, sympatric speciation occurs in populations that are not geographically isolated.

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        geographic isolation
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That's correct.

Correct! If you had trouble with this question, review the following material: Allopatric speciation occurs when a population becomes geographically isolated from other members of its species; in contrast, sympatric speciation occurs in populations that are not geographically isolated. Polyploidy, sexual selection, and natural selection can all promote speciation, but they can do so in sympatric populations as well as allopatric populations.

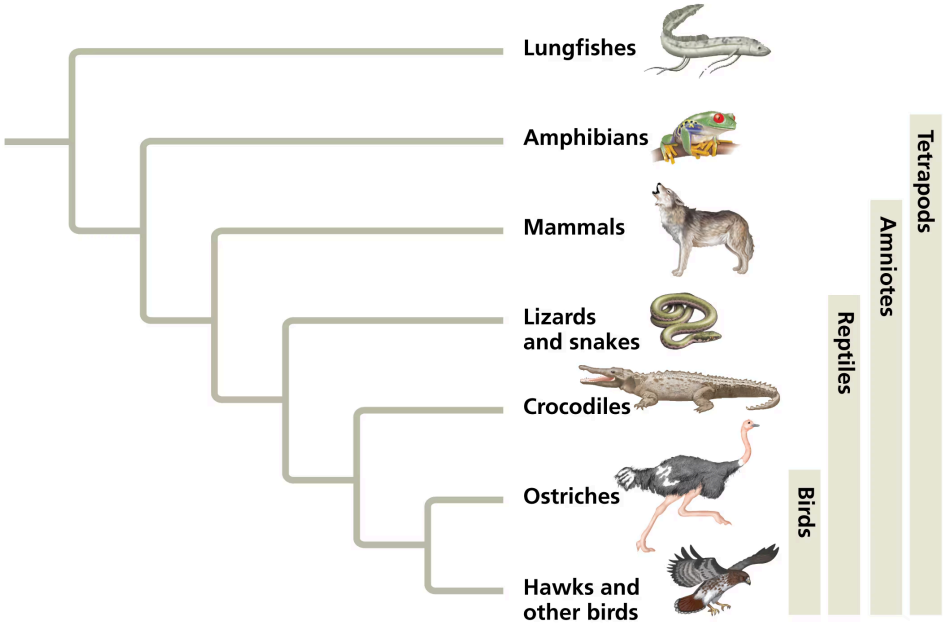

Diagram showing a series of horizontal forked branches. From the left, the first branch point forks into a horizontal line leading to Lungfishes and another horizontal line, which branches again. This second branch point forks into a horizontal line leading to Amphibians and another horizontal line, which branches again. This third branch point forks into a horizontal line leading to Mammals and another horizontal line, which branches again. This fourth branch point forks into a horizontal line leading to Lizards and snakes and another horizontal line, which branches again. This fifth branch point forks into a horizontal line leading to Crocodiles and another horizontal line, which branches a final time into two groups, Ostriches at the end of the top branch and Hawks and other birds at the end of the bottom branch. At the far right of the diagram, a vertical bar labeled Birds spans these two branches. Another vertical bar labeled Reptiles spans the Birds and also the Crocodiles and Lizards and snakes branches. Another vertical bar labeled Amniotes spans the Reptiles and also the Mammals branches. Another vertical bar labeled Tetrapods spans the Amniotes and also the Amphibians branch.

3. Based on the evolutionary tree shown, mammals share a most recent common ancestor with which groups?

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lizards and snakes, crocodiles, and birds
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That's correct.

Correct! If you had trouble with this question, review the following material: Mammals share a more recent common ancestor with lizards and snakes, crocodiles, and birds than they do with amphibians. This is indicated by the branch point that splits into the branches leading to the mammals and to the lizards and snakes, crocodiles, and birds, which is located to the right of the branch point that represents the most common ancestor of amphibians and the mammal and reptile lineages. Hence, mammals are more closely related to lizards and snakes, crocodiles, and birds than they are to amphibians. Note also that mammals are as closely related to lizards and snakes as they are to crocodiles or birds.

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amphibians and lizards and snakes
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That's incorrect. Try again.

Incorrect. Review the following material: Mammals share a more recent common ancestor with lizards and snakes, crocodiles, and birds than they do with amphibians. This is indicated by the branch point that splits into the branches leading to the mammals and to the lizards and snakes, crocodiles, and birds, which is located to the right of the branch point that represents the most common ancestor of amphibians and the mammal and reptile lineages.

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amphibians
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That's incorrect. Try again.

Incorrect. Review the following material: Mammals share a more recent common ancestor with lizards and snakes, crocodiles, and birds than they do with amphibians. This is indicated by the branch point that splits into the branches leading to the mammals and to the lizards and snakes, crocodiles, and birds, which is located to the right of the branch point that represents the most common ancestor of amphibians and the mammal and reptile lineages.

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lizards and snakes
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That's incorrect. Try again.

Incorrect. Review the following material: Mammals share a more recent common ancestor with lizards and snakes, crocodiles, and birds than they do with amphibians. This is indicated by the branch point that splits into the branches leading to the mammals and to the lizards and snakes, crocodiles, and birds, which is located to the right of the branch point that represents the most common ancestor of amphibians and the mammal and reptile lineages.

Learning Objectives

After studying this chapter, you should be able to do the following:

- 25.1Describe steps by which simple cells may have originated from nonliving materials.
- 25.2Explain what fossils are, how they are dated, and what the fossil record can reveal about life's history.
- 25.3Identify when the origin of single and multicelled organisms and the colonization of land occurred, and explain the significance of these events.
- 25.4Explain how plate tectonics, mass extinctions, and adaptive radiations have affected Earth's life.
- 25.5Describe how changes in the sequence or regulation of genes can result in major changes in body form.
- 25.6Use examples to show how novel and complex structures can arise by descent with modification.